Legal Research and Evidence Assisstant-“JuriAid”

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Proposal Document

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July 2025

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Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Abstract

The Sri Lankan judicial system is hampered by outdated legal research methods, which take time. Retrieval of legal documents is often automated, but there are no advanced algorithms capable of understanding the full context of the documents and producing meaningful outputs. Solving this problem is challenging as most solutions do not account for the unique legal system of Sri Lanka.

This project proposes to develop a new system to find past court decisions similar to a new court decision based on their similarities using a knowledge graph. The system will use computational models to automatically classify legal document lexemes into basic logical roles (e.g., facts, issues) and obtain similarity through a weighted hybrid similarity metric based on these parts. The system will integrate citation links and their semantic significance into the knowledge graph to enable multi-hop reasoning to find indirect connections between cases. The system will be tailored to the Sri Lankan context, allowing for direct PDF uploads, English text processing, and interpretable outputs. The system seeks to improve the efficiency, accuracy, and accessibility of the Sri Lankan legal system.

Keywords: Legal Knowledge Graphs, Semantic Similarity, Rhetorical Role Identification, MultiHop Reasoning, Computational Linguistics, Sri Lankan Law, Case Similarity, Data-Driven Legal Tech.

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1.Introduction

In Sri Lanka, the legal system follows the doctrine of precedent, where earlier rulings guide how laws are interpreted and applied. lawyers rely on these decisions to support arguments and shape judgments in new cases.

However, finding and analyzing judgments is still slow and complex. Existing digital databases rely mainly on keyword searches, often missing cases that use different wording for similar reasoning.

To address this, the Past Case Retrieval Component is being developed within the Legal Research and Evidence Assistant system. It integrates NLP, Knowledge Graphs, Graph Neural Networks (GNNs), and Retrieval-Augmented Generation (RAG) to uncover precedents beyond direct citations or identical language.

A key feature is classifying sentences by rhetorical roles—Facts, Issues, Arguments, and Decisions—an approach shown by Sheetal et al. (2022) to improve case similarity over full-text or citation-based methods [1]. Knowledge Graphs further strengthen this by mapping legal entities (cases, statutes, judges, principles) and their relationships, enabling contextual understanding (Fensel et al., 2020) [2]. Such graphs have already enhanced retrieval, question answering, and recommendations in legal domains [3].

GNNs build on this by learning from graph structures to predict hidden links like case similarity or citations. Dhani et al. (2024) demonstrated that RGCNs are effective for Indian judgments when combined with expert-defined “law points” [3]. Similarly, this system will apply GNNs to Sri Lankan case law.

Finally, RAG grounds outputs in real precedents, ensuring that generated summaries and recommendations are both accurate and traceable.

Overall, the system aims to modernize legal research in Sri Lanka—making it faster, more precise, and transparent. By combining explainable AI with precedent-based reasoning, it will not only locate the most relevant cases but also justify their selection, ensuring both efficiency and trust.

2.Literature Review

Legal information retrieval has traditionally relied on keyword-based and citation-based search methods, which, while effective for exact matches, often fail to capture deeper semantic and contextual relationships between cases [4]. In the legal domain, especially for precedent-based systems such as in Sri Lanka, the ability to find cases with similar legal reasoning rather than identical terminology is critical.

One approach to improve retrieval effectiveness is rhetorical-role segmentation, where judgments are divided into functional sections such as *Facts*, *Issues*, *Arguments*, *Analysis*, and *Decision* [5]. Hierarchical sequence labeling models, such as SciBERT-HSLN, have been shown to accurately classify sentences into these roles, allowing retrieval systems to compare specific sections (e.g., *Facts-to-Facts*) rather than entire documents [6]. This method has consistently outperformed traditional document-levael similarity in legal contexts [7].

Research also highlights the benefits of role-aware similarity scoring, where textual similarity is calculated within comparable rhetorical segments. TF-IDF combined with cosine similarity has been found to yield strong performance, and giving higher weight to legally significant sections, such as *Facts* and *Analysis*, further improves retrieval accuracy [8].

Beyond text-based methods, Knowledge Graphs (KGs) have emerged as powerful tools for representing legal cases and their relationships [9]. In a KG, cases, statutes, legal concepts, and entities (e.g., parties, courts) are represented as interconnected nodes, with edges denoting relationships such as citations, legal principles, or case similarity. This graph-based representation allows for both efficient query execution and explainable results, as relationships between cases can be visualized [10].

Recent advances in Graph Neural Networks (GNNs), such as Relational Graph Convolutional Networks (R-GCN), have been applied to legal knowledge graphs for tasks like citation prediction and similarity-based ranking [11]. However, studies indicate that the performance of these models depends heavily on the availability of domain-specific features, such as legal concept annotations or local embeddings, which are currently lacking for Sri Lankan case law [12].

Retrieval-Augmented Generation (RAG) has also been explored in the legal domain to integrate semantic retrieval with natural language generation. In such systems, a vector database (e.g., FAISS, Pinecone) stores embeddings of legal documents, which are retrieved using models like Sentence-BERT or Legal-BERT [13]. The retrieved cases are then passed to a language model (e.g., GPT-4, LLaMA) to produce summaries or explanations grounded in the retrieved sources. While RAG primarily serves question answering or drafting, its retrieval component can be adapted directly for similar-case search [14].

Despite these advances, most existing work has focused on jurisdictions with publicly available datasets, such as the United States or India [15]. There is a lack of publicly accessible, structured, and annotated Sri Lankan legal corpora, especially those with rhetorical role segmentation.

Developing such resources would enable the adaptation of these state-of-the-art methods to the Sri Lankan context, improving both the accuracy and explainability of similar case retrieval.

3.Research Gaps

1. No Sri Lankan-Specific Legal Datasets Most existing legal AI systems rely on well-structured datasets from countries like India, the US, and Europe. Unfortunately, Sri Lanka doesn't yet have a publicly available, digitized, and properly organized collection of court judgments. This makes it difficult to train or test modern legal retrieval systems for local use.

2. Missing Role-Based Annotations in Sri Lankan Cases Breaking down judgments into structured sections—like Facts, Issues, Arguments, and Decisions—has been proven to improve case retrieval. However, no such annotated dataset exists for Sri Lankan case law. Without these labels, AI models can't compare cases based on their legal reasoning—only on superficial keyword matches.

3. Knowledge Graphs Haven't Been Applied to Sri Lankan Law Knowledge Graphs (KGs) are powerful tools for mapping connections between cases, laws, and legal concepts. While they've been used successfully in other countries, no one has yet built a Sri Lankan legal KG that combines case citations, semantic similarity, and role-based features.

4. Retrieval-Augmented Generation (RAG) Remains Untested for Sri Lankan Precedents RAG systems—which combine AI-powered search with natural language generation—are being used in places like the US and India for legal research. But no one has yet adapted this approach for Sri Lanka, especially not in a way that integrates role-based retrieval and knowledge graphs.

5. Most Systems Ignore PDF Processing—A Huge Practical Barrier Many legal AI tools assume clean, pre-extracted text. But in reality, lawyers and researchers work with PDFs—scanned documents, court filings, and unstructured case files. Without built-in PDF parsing, users have to manually extract text before running searches. This project solves that problem by allowing direct PDF uploads and automatic text processing for Sri Lankan cases..

4.Research Problem

Although legal case retrieval systems have shown some effectiveness in other countries, they perform poorly within the Sri Lankan legal context due to several key limitations. Most existing tools rely primarily on keyword matching, which fails to capture deeper semantic relationships in legal reasoning. Additionally, these systems overlook the rhetorical structure of judgments, treating them as unstructured text rather than differentiating between sections such as Facts, Issues, and Reasoning, which are crucial for accurate case similarity assessments.

Current retrieval approaches also tend to separate semantic similarity from citation networks, rather than integrating both, which limits their ability to uncover richer and more meaningful connections across precedents. Moreover, many AI-driven retrieval systems lack transparency, offering little explanation as to why a case was recommended—reducing trust and usability among legal practitioners. Finally, while knowledge graphs hold promise for capturing complex legal relationships, their rapid growth often leads to scalability and efficiency challenges, making them difficult to apply effectively in large collections of Sri Lankan case law.

5.Objectives

Main Objective

To develop a hybrid knowledge graph and vector database-driven retrieval system capable of input Sri Lankan court case PDFs and retrieving similar past cases with role-wise explainabilityB. Specific

Specific Objectives

1. Read Cases Directly from PDFs

Lawyers and judges work with PDFs — scanned documents, court filings, judgments. Right now, most AI tools can't handle them directly.

This system will let users upload a PDF of a Sri Lankan court case, and it will automatically pull out the text — cleanly and accurately — using reliable tools like PyMuPDF or pdfminer.

2. Break Down the Case Into Its Key Parts

Every legal judgment tells a story with structure:

What happened? → Facts

What's being disputed? → Issues

What did each side say? → Arguments

What was the final decision? → Decision

The system uses smart natural language processing (NLP) to automatically sort every sentence into these roles. This helps it understand the case like a lawyer would — not just as a block of text, but as a structured legal argument.

3. Build a Smart Legal Network

Instead of treating past cases like entries in a long list, this system organizes them like a web of connected legal knowledge think of it as a mind map for the law, built using Neo4j.

Each case is a node in the network.

4. Find Cases That Are Truly Similar — Even If They Use Different Words

It's not enough to match keywords. Two cases might talk about "unauthorized use of property" and "trespassing" different words, same idea.So the system uses smart AI models like Legal-BERT or Sentence-BERT to turn each part of a case into a meaning-based summary (embedding).These are stored in a vector database (like FAISS or Pinecone) so the system can quickly find cases that are semantically similar even if they don't share the same phrases.

5. Show Why a Case Was Recommended

AI shouldn't be a black box. If the system suggests a past case, you should know why.

So it doesn't just return a list — it explains:

6.Methodology

The development of this AI-powered legal question generation system will follow a systematic methodology, encompassing data collection, model development, system integration, and rigorous

testing. The overall system aims to provide a comprehensive solution for legal question automation.

Overall System

The whole system, as described by your context, can be conceptualized as an AI-powered legal assistant with a core component for question generation. Referring to the provided system diagram, the overall architecture of such a legal AI system would include:

- User Input: Lawyers provide initial case details, complaints, and replies [User's query, 135].
- Agent Framework Orchestrator: Manages task distribution and overall workflow.
- Court Decision Retrieval & Law Retrieval Engine: These components are responsible for fetching relevant statutes, precedents, and case law from a comprehensive legal knowledge graph (e.g., Neo4j) and embedding stores (using FATSS/SBERT and embeddings).
- Questionnaire Generator (Core Component): This module, powered by fine-tuned Legal-BERT with LoRA adaptation, dynamically generates context-aware questions and assesses legal positions.
- Legal Position Evaluation & Report Generator: Assesses the legal position based on input and generated questions, leading to output synthesis (e.g., draft questions, summaries).
- Long-Term Memory: Stores query history and aids in continuous learning.
- Final Output: Comprehensive legal analysis, guidance, and actionable recommendations (in your case, the generated legal questions).

OVERALL SYSTEM DIAGRAM

Figure 6.1 System Overview Diagram

Individual Component

This system helps lawyers quickly find past cases similar to a new one — not by keyword, but by meaning and legal structure. It's built for real use: upload a PDF, get smart, explainable results.

System work like,

1. Upload & Read PDFs

Start with a real court case PDF. The system pulls out the text (using PyMuPDF or pdfminer), cleans up junk like headers, and splits it into sentences.

2. Break It Down Like a Lawyer

Every sentence is tagged as Facts, Issues, Arguments, or Decision — using smart NLP models like Legal-BERT and SciBERT-HSLN. This helps the system understand the case the way a lawyer does.

3. Build a Legal Network (Knowledge Graph)

Using Neo4j, the system maps out cases, laws, parties, and how they're connected — like "this case cited that one" or "both deal with trespass." It's a web of legal knowledge.

4. Turn Text into Meaning (Vectors)

Key sections are converted into semantic vectors using Sentence-BERT and stored in a fast vector database (FAISS or Pinecone). That way, it can find cases that mean the same thing — even if they use different words.

5. Find Matches & Explain Why

When you search, the system finds similar cases using both meaning and citations. It shows why they match — like "Facts are 90% similar" — and highlights the relevant parts. A smart AI (like GPT-4) then summarizes the results in plain language.

6. Ready to Use

The backend runs on FastAPI/Flask, the frontend is built with React.js, and everything's packaged in Docker — so it works in the cloud (AWS/Azure) or on a local server.

INDIVIDUAL COMPONENT SYSTEM DIAGRAM

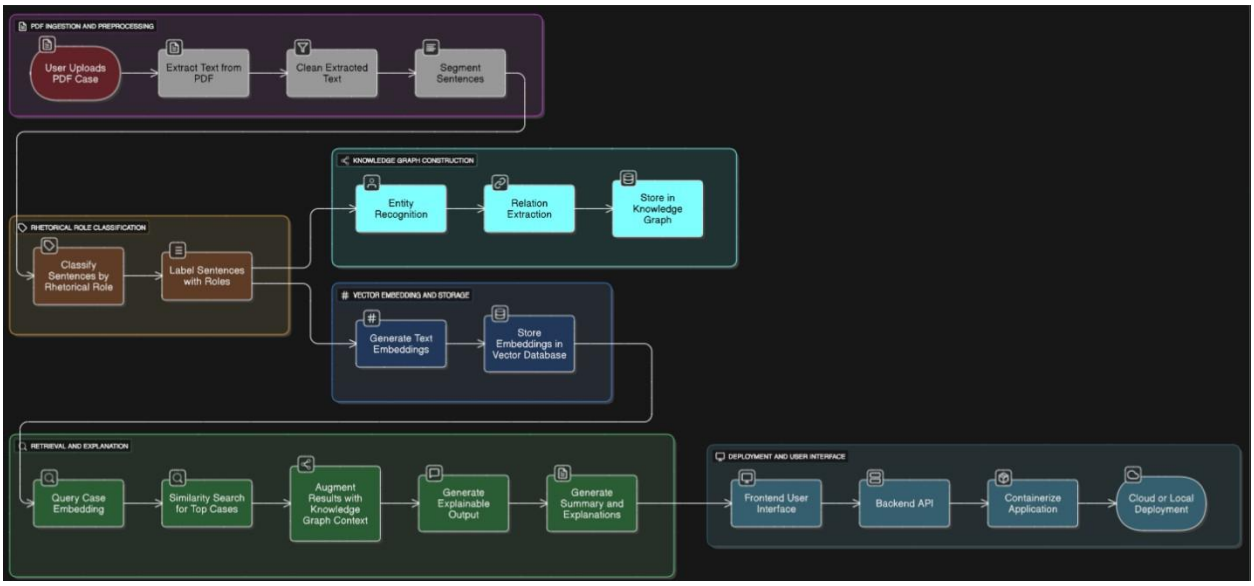

Figure 6.2 Component system diagram

7.Project Requirements

The system requirements for the Past Case Retrieval component are divided into two categories: Functional Requirements and Non-Functional Requirements.

A. Functional Requirements

The functional requirements define the core capabilities and operations that the system must perform.

1. PDF Case Upload – The system shall allow users to upload Sri Lankan court case documents in PDF format.
2. Text Extraction – The system shall extract text from PDF files using document parsing tools.
3. Text Preprocessing – The system shall remove headers, footers, and unwanted formatting, and segment text into sentences.
4. Rhetorical Role Classification – The system shall classify sentences into predefined roles such as Facts, Issues, Arguments, and Decision.
5. Knowledge Graph Construction – The system shall store extracted entities and relationships in a Neo4j-based knowledge graph.
6. Vector Embedding Generation – The system shall generate dense vector representations of legal text segments.
7. Vector Database Storage – The system shall store embeddings in a similarity search database (e.g., FAISS or Pinecone).
8. Case Similarity Retrieval – The system shall retrieve the top-N most similar past cases based on vector similarity.
9. Knowledge Graph Augmentation – The system shall enrich retrieval results with linked legal entities and citations.
10. Explainable Output Generation – The system shall present role-wise similarity scores, matched sections, and reasoning traces.
11. Summarization – The system shall generate natural language summaries of retrieved cases using an LLM within the RAG framework.
12. User Interface – The system shall provide a web-based interface for case upload, retrieval, and visualization.

B. Non-Functional Requirements

The non-functional requirements define the quality attributes and constraints for the system.

1. Performance – The system shall return retrieval results within 5 seconds for typical query cases.
2. Scalability – The system shall handle up to 10,000 stored case documents without significant performance degradation.
3. Accuracy – The rhetorical role classification model shall achieve at least 85% F1-score on the test dataset.
4. Reliability – The system shall ensure 99% uptime during operational hours.
5. Security – Uploaded documents shall be stored securely, with access restricted to authorized users.
6. Maintainability – The system architecture shall support modular updates for NLP models and database components.
7. Usability – The web interface shall be intuitive and require no more than 10 minutes of user training.
8. Portability – The system shall be deployable on both cloud infrastructure (AWS/Azure) and local servers using Docker.
9. Compatibility – The system shall be compatible with major web browsers (Chrome, Firefox, Edge).
10. Data Privacy – The system shall comply with Sri Lankan legal data privacy and confidentiality standards.

8.Commercialization

Budget

The following amounts are estimated and may be subject to change:

Description	Cost	Occurring
Server Hosting	**LKR 15,000**	**Monthly**
Cloud Database	**LKR 20,000**	**Monthly**
Developer Tools/Software Subscriptions	**LKR 5,000**	**Monthly/Annual**
LLM API usage	**LKR 10,000**	**Monthly**

Table8.1budgettable

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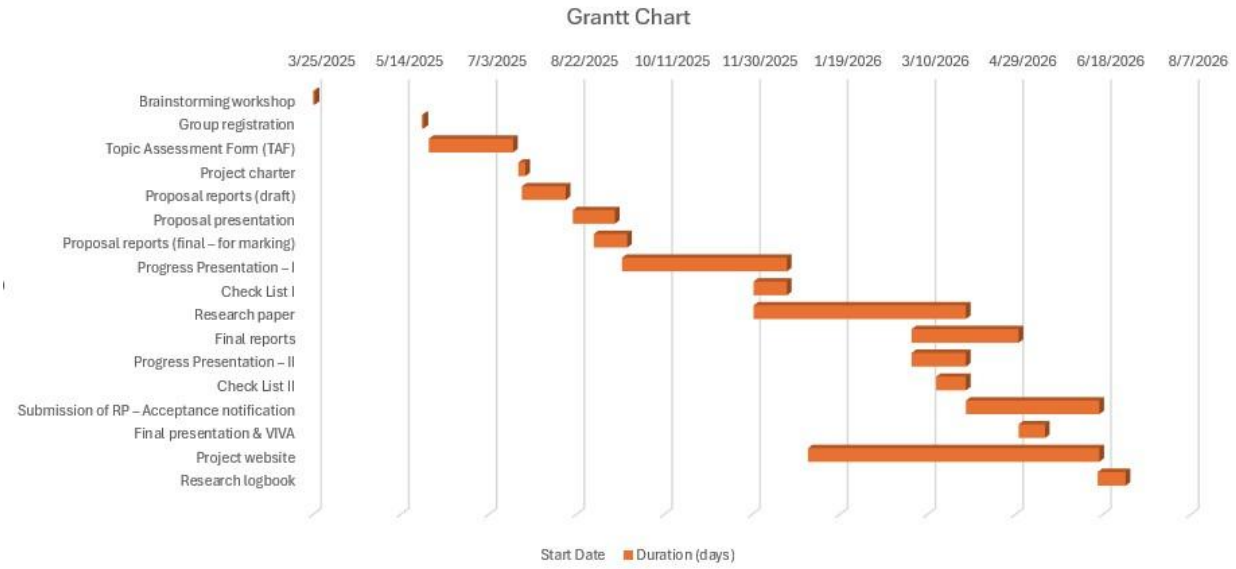
APPENDICES

Appendix 1. Gantt Chart



JURIAID – AN AI-POWERED LEGAL RESEARCH AND EVIDENCE ASSISTANT

Project ID : 25-26J-274

Project Proposal Report

Prarthana A.P.S – IT22127228

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Co-supervised by: Ms. Madusha Weerasooriya

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July 2025

DECLARATION

We are confirming that the work given in this proposal is one of our own volition. The same work has not been given, entirely or partially, for academic credit at any other university or institution. We have also made sure that all sources of information and ideas from others have been clearly cited and appropriately acknowledged in the document.

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The above candidates are researching the undergraduate Dissertation under my supervision.

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Signature of co-supervisor: Date: 2025/08/29

ABSTRACT

This project removes a serious inefficiency of Sri Lankan legal practice: the time-consuming, manual process of looking up, verifying, and placing in context relevant statutes and amendments to them. Present legal databases offer only skeletal keyword search, have no semantic understanding, and perform no automatic follow-through on law changes, and therefore force lawyers to devote 6–10 hours per case to statutory research. Our proposed solution is LawStatKG, which is an AI-based legal retrieval system that streamlines and automates statute identification. It extracts relevant Sri Lankan statutes and amendments through hybrid retrieval (BM25 + LegalBERT embeddings), a legal knowledge graph (RDF/Neo4j), and temporal tracking of amendments. It accepts natural language English queries, retrieves conceptually related law, and brings out automatically all relevant amendments and cross-references. The core design employs LegalBERT optimized for semantic understanding, a legal domain-specific RDF ontology to model legal relations, and a version-aware timeline system to track the provenance of laws. This enables the system to offer courtroom-ready, rolled-up output with the statute text, amendment history, and referenced provisions in one view. By automating the essential legal task of fetching statutes, the research aims to reduce case preparation time significantly, decrease legal errors, and make access to accurate statutory information democratised. The system emphasizes reliability, usability, and interworking with the common AI legal assistant platform to render it realistically deployable in the Sri Lankan legal environment.

Keywords: Legal AI, Statute Retrieval, Hybrid Search, Legal Knowledge Graph, Temporal Tracking, Legal-BERT, Legal Tech, Sri Lanka

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LIST OF ABBREVIATIONS

ABBREVIATION	DESCRIPTION
LawStatKG	Law Statute Knowledge Graph
KG	Knowledge Graph
LLM	Large Language Model
BM25	Best Match 25
CTV	Component Temporal Version
TV	Temporal Version
TF-IDF	Term Frequency-Inverse Document Frequency
NLP	Natural Language Processing
e-Laws	Electronic Laws Portal
SPARQL	SPARQL Protocol and RDF Query Language
NLL	Natural Language Leagal

1. INTRODUCTION

1.1 BACKGROUND

Punctuality, accuracy, and scale of statutory research in modern legal practice directly impact the quality of legal representation and administration of justice. Legal researchers, lawyers, and judges entirely depend on timely and accurate availability of relevant laws, amendments, and associated statutes during preparation of legal arguments or judicial decisions. Delayedness or inaccuracy in identification of provisions of law that are applicable can lead to weakening of legal submissions, procedural delay, or denial of justice. Statutory research in Sri Lanka is primarily, manual, and laborious. The process typically consists of initial search for the main legislation that regulates a case, and subsequent comprehensive verifications of its present validity, applicable amendments, repeals, and cross-references to related acts. Lawyers usually rely mainly on the e-Laws portal, gazettes, and bulk PDF repositories. These machines tend to operate almost completely on word-level searches, which tend not to capture the underlying legal intent or word-level nuances. Second, amendments also have to be complied with manually by cross-verifying different sources, and there is no facility of an automatic system encompassing all the content incorporating its reasonable integration. It is a labor-intensive process, prone to leaving something out, and particularly inconvenient for rural practitioners who have no easy access to up-to-date legal materials and networks of expertise. The absence of semantic understanding, relationship mapping, and history tracking in existing legal research tools is thus evidently a technology deficit.

This is especially difficult in a common law jurisdiction like Sri Lanka's, where statutory interpretation requires not only consideration of the content of the statute itself but also of its history, context, and relationship to other statutes. In reality, this means lawyers taking between six to ten hours to prepare for a single case, with corresponding high probabilities of failing to catch relevant amendments or interfacing provisions, particularly for complex legal issues. This research proposes LawStatKG, a legal statute knowledge graph module with AI integrated into a larger legal assistant software.

Distinct from conventional search engines, LawStatKG combines semantic retrieval, relation-aware navigation, and temporal versioning to provide an ever-changing map of Sri Lankan law. It will recognize natural language questions in law, import relevant statutes using a hybrid approach combining lexical search through BM25 with searching by semantic embeddings using LegalBERT, and link each statute to its amendments and related acts into a knowledge base specifically adapted to RDF. The system will also track historical development of provisions in law, showing how laws have developed over time and when each amendment became active, and constructing aggregated, attorney-level reports that combine contemporary statutory language, pertinent amendments, and coterminous laws into one last authoritative reference. By transforming isolated legal knowledge into a smart, consolidated, and searchable

set of information, LawStatKG will cut research time by a significant percentage, equalize access to good quality legal information, and make case preparation more accurate and more so. It will bridge the urban–rural divide in legal research capability and help build Sri Lanka's current legal information infrastructure. Lastly, the aim of this study is to show how these synergies between semantic search, knowledge graph technologies, and natural language processing can make individuals more accessible to justice, enhance the practice of law, and pay long-term dividends to the legal community.

1.2 LITERATURE SURVEY

The use of Artificial Intelligence in legal information systems has progressed significantly with the evolution of Retrieval-Augmented Generation (RAG), a system that harmonizes external knowledge retrieval with the generative capabilities of Large Language Models (LLMs). This is consistent with how it addresses the inherent vulnerabilities of standalone LLMs—such as hallucinations, out-of-date information, and untraceable reasoning—by anchoring answers to verifiable, external proof, hence making it highly suitable for high-risk applications like law [1].

Hybrid Retrieval: Merging Different Search Paradigms for Robustness

Yet another crucial advancement in RAG systems is the implementation of hybrid retrieval, when different paradigms of retrieval are brought together to offer more robustness and accuracy. While Gao et al. present a comprehensive review of RAG advancements, they note that naive retrieval (BM25 or dense embeddings) generally fails to achieve a balance of precision and recall in complex domains [1]. To oppose this, Kalra et al. propose HyPA-RAG, a system that integrates sparse (BM25), dense (embedding-based), and knowledge graph (KG)-based retrieval into a single pipeline [2]. This combination allows the system to utilize keyword matching, semantic semantics, and relational reasoning simultaneously. The effectiveness of hybrid retrieval is also confirmed by Sheetal et al., who experiment with TF-IDF, BM25, and Sentence-BERT for legal document similarity, noticing that while each method has its strengths, their combination yields more consistent results [3]. Similarly, Silva et al. demonstrate in Graph RAG how the retrieval performance is boosted when the structural as well as semantic information are merged, although they don't use BM25 [4]. These findings collectively suggest that there is no one retrieval strategy that will effectively work for legal AI, and that a hybrid multi-strategy approach is required to obtain reliable results.

Knowledge Graphs: Enabling Relational and Contextual Reasoning

Knowledge Graphs (KGs) in RAG systems have now become the first-order innovation for contextual intelligence improvement. Kalra et al. provide a KG of policy objects (requirements, obligations) and introduce KG triplets as additional context to the LLM, improving traceability

and explainability of outputs [2]. This allows the system to reason about legal concept relationships instead of isolated text.

Sheetal et al. follow the same path by developing a Neo4j-based KG of Indian court judgments, in which rhetorical roles (Facts, Ruling, Precedent) are nodes and relation between them is edges [3]. Their study demonstrates that the graph structure of legal text allows for thematic similarity computation and makes case law recommendation systems possible. Silva et al. extend this idea in Graph RAG, where legal knowledge graph is not only a fixed structure but a dynamic, hierarchical graph that includes legal norms, components, and versions [4]. It allows for structure-aware retrieval where queries can be targeted to certain levels (Article, Section) of a statute. Collectively, these works demonstrate that KGs are not repository software but dynamic entities that augment retrieval, reasoning, and response generation in legal AI.

Temporal and Structural Modeling: Modeling Legal Evolution

Legal texts are dynamic; they evolve over time by repeals, amendments, and revisions. Silva et al. address this in Graph RAG by introducing Temporal Version (CTV) nodes, a snapshot of a legal provision at a particular point of time [4]. This enables point-in-time queries critical property for correct legal interpretation. The model also handles alteration actions ("amended by Act Z") as first-class things, so the system has a record of the history in which the laws evolved. Even though Gao et al. and Kalra et al. do not model temporal change directly, their designs do support versioned data integration in retrieval and augmentation processes [1], [2]. Neither Sheetal et al. supports temporal tracking because they're more concerned with judgment similarity than statutory change [3]. The absence of temporal modeling in most systems, however, highlights the innovation of Graph RAG in addressing the dynamic nature of law.

Though these four papers are a huge leap forward—Gao et al. [1] extending RAG, Kalra et al. [2] suggesting hybrid retrieval, Sheetal et al. [3] employing KGs to model legal judgments, and Silva et al. [4] advancing structural and temporal modeling—none of them are for Sri Lankan law. There is no system that combines hybrid retrieval, a Sri Lankan statute legal knowledge graph, automated amendment linking, and temporal changes tracking into an end-to-end pipeline yet.

This work fills that gap by presenting LawStatKG, the first Sri Lanka-centered legal retrieval engine that benefits from the strengths of these approaches. By hybrid retrieval over HyPA-RAG, KG methods of Indian legal NLP, and temporal-structural graph modeling from Graph RAG, the system presented here offers a valid, context-tailored, and timely answer for statutory research in Sri Lanka.

1.3 RESEARCH GAP

Despite the incredible advancements in Retrieval-Augmented Generation (RAG) and its implementation in legal applications, there remains one overwhelming research gap: the absence of a hybrid, knowledge-driven, time-sensitive legal retrieval system suitable for the Sri Lankan statutory framework. While the four said works—Gao et al. [1], Kalra et al. [2], Sheetal et al. [3], and Silva et al. [4] - collectively represent the state of the art in RAG for legal purposes, none of them takes into account the pertinent structural, linguistic, and procedural characteristics of Sri Lankan law, and there is no existing system that encompasses all building blocks needed to facilitate proper, context-dependent, and historically informed legal research in the jurisdiction. A look at the literature reveals that key capabilities have been developed in isolation but not yet integrated for Sri Lanka.

To identify the research gap, we compared four research papers in more recent works to our solution. The results are shown in Table 1.3.1.

Feature	Research (Y.Gao) [1]	Research (R.Kalra) [2]	Research (S.sheetal) [3]	Research (J.silver) [4]	Proposed System
Hybrid Retrieval (Combines TF-IDF keyword search and embeddings)	Discussed. But not implement	Yes	No	Partial	Yes
Legal Knowledge Graph	No	Yes	Yes	Yes	Yes
Automated Amendment Linking	No	No	No	Yes	Yes
Temporal Tracking of Legal Changes	No	No	No	Yes	Yes
Jurisdiction-Specific Design	No	Yes- USA	Yes- India	Yes - Brazile	Yes -Sri lanka
Combine System	No	No	No	Yes	Yes

Table 1.3.1 Comparative analysis highlighting the novelty of the proposed system

1. Hybrid Retrieval is Proven, but Not Yet for Sri Lankan Law

The hybrid retrieval concept—combining sparse (e.g., BM25), dense (e.g., embeddings), and knowledge-based methods—is proven. Gao et al. [1] summarize RAG's development and assert the superiority of hybrid over one-method retrieval. Kalra et al. [2] demonstrate this in practice with HyPA-RAG, which integrates BM25, semantic embeddings, and knowledge graph triplets to facilitate higher accuracy and prevention of hallucinations in legal systems for AI. However, their system is designed for NYC Local Law 144, not Sri Lankan legislation. While the method is established, it has not been applied to the Commonwealth-form legislative code, multilingual setting (English, Sinhala, Tamil), or Sri Lankan juridical vocabulary.

2. Knowledge Graphs Add Context, but Aim Case Law, Not Statutes

Using knowledge graphs (KGs) to harvest legal relations is a familiar refrain. Sheetal et al. [3] construct a Neo4j-based knowledge graph (KG) of Indian Supreme Court judgments, where rhetorical roles (e.g., Facts, Ruling, Precedent) are nodes and thematic similarity between nodes are edges. This enables more intelligent case recommendations and similar computation. Kalra et al. [2] also uses a KG to identify relevant policy entities and inter-entities. Both systems are interested in judicial rulings or policy compliance, not statutory law. There is no KG that represents Sri Lankan Acts, Sections, Amendments, and their interconnection - a primary need for statutory research.

3. Temporal and Structural Modeling is Advanced but Geographically Limited

The most impactful structural and temporal modeling innovation is attributed to Silva et al. [4], who propose Graph RAG. This method depicts legal norms as a hierarchical and temporal graph with Component Temporal Version (CTV) nodes containing the text of a provision at a time. This facilitates point-in-time queries - a fundamental requirement for determining the law on a previous date. Their contribution is novel, albeit experimented upon the Brazilian legislation system, which has a unique legislative drafting style from the one in Sri Lanka based on the Commonwealth. The technique is sound, but not in the local mode.

4. No Integrated Sri Lankan Statutory Research System

Current tools made available to Sri Lankan legal practitioners, e.g., the e-Laws portal, offer only rudimentary keyword searching without:

Semantic understanding of legal concepts,

Cross-referencing of amendments done automatically,

Relationship mapping between enactments,

Or historical version tracking.

1.4 RESEARCH PROBLEM

The main problem this research attempts to address is that there does not exist any reliable way for Sri Lankan lawyers and legal researchers to quickly and accurately retrieve the current or past version of the law. Sri Lankan lawyers, judges, and legal researchers now must check manually at the time of the current moment and past version of laws. They rely on the e-Laws portal, which supports only rudimentary keyword searching and lacks automated provisions to monitor amendments, merge updated provisions or find the very specific wording of a law as of a specific date.

While the current hybrid retrieval systems have made the integration of semantic search, knowledge graphs, and temporal tracking more likely in legal analysis, such systems are available in the foreign jurisdictions (USA, Brazil) and are not available or implemented for Sri Lanka law. This paper aims to overcome this limitation by developing LawStatKG, the first legal retrieval engine specifically designed for Sri Lanka that combines:

1. Hybrid retrieval (BM25 + LegalBERT) for precise search.
2. A legal knowledge graph to map the relationships among amendments and laws,
3. Temporal versioning to follow the evolution of the laws over time, And automated combination of current and past law books.

RESEARCH QUESTION

There does not exist any reliable way for Sri Lankan lawyers and legal researchers to quickly and accurately retrieve the current or past version of the law.

2. RESEARCH OBJECTIVES

2.1 MAIN OBJECTIVE

To create and implement LawStatKG, a hybrid, knowledge-based, and temporal legal search environment for Sri Lankan statute law. The environment will include semantic search, legal knowledge graph, and temporal versioning to allow practitioners to automatically find up-to-date and historical laws, remove manual labor, remove errors, and improve access to accurate legal information throughout Sri Lanka.

2.2 SPECIFIC OBJECTIVES

1. **To develop a hybrid retrieval engine that incorporates keyword-based, semantic, and knowledge graph-based methods for precise legal search.** This is achieved by constructing a retrieval pipeline that integrates BM25 for robust keyword match, LegalBERT embeddings for understanding legal term meaning, and a knowledge graph (KG) retriever to extract suitable legal relationships. The results of the three methods will be combined by Reciprocal Rank Fusion (RRF) to produce a better and more comprehensive result set than any single approach alone.

2. **To construct a Sri Lanka-specific legal knowledge graph (LKG) that would replicate the hierarchical structure of statutes and their interrelations.** This involves building an RDF/Neo4j-powered graph in which legal entities such as Acts, Parts, Chapters, and Sections are considered nodes, and relationships between them (amends, repeals, cites) as edges. The graph will enable "semantic navigation" to allow users to travel from a general subject to a specific clause in legislation, such that the context returned is comprehensive according to the law.

3. **To enable a temporal versioning system capturing the evolution history of modifications in the legislation over time.**
   That is, every legal provision will be represented using Temporal Version (CTV) nodes, which contain a precise copy of a law at a specific date. This framework will automatically propagate, so that whenever an update is done to a section, all its parent pieces (articles, chapters) that include it get updated to a fresh version. This is done by generating Text Units for each legal provision, including not just the isolated text of an individual section, but also the necessary context of its parent pieces (the capture of an article). The Text Units will be used for retrieval and provided to the LLM for the

purpose of generating a final output with the statute text, its amendment history, and surrounding provisions, in a human-readable, easy-to-understand format.

3. METHODOLOGY

Development of LawStatKG will be performed employing a systematic approach with five broad phases: Data Acquisition, preprocessing, model construction, knowledge graph construction, and system integration and testing. The approach will combine Natural Language Processing (NLP), semantic search techniques, and knowledge representation models to deliver a time-critical, relationship-sensitive statute retrieval system tailored to Sri Lankan law.

OVERALL SYSTEM DIAGRAM

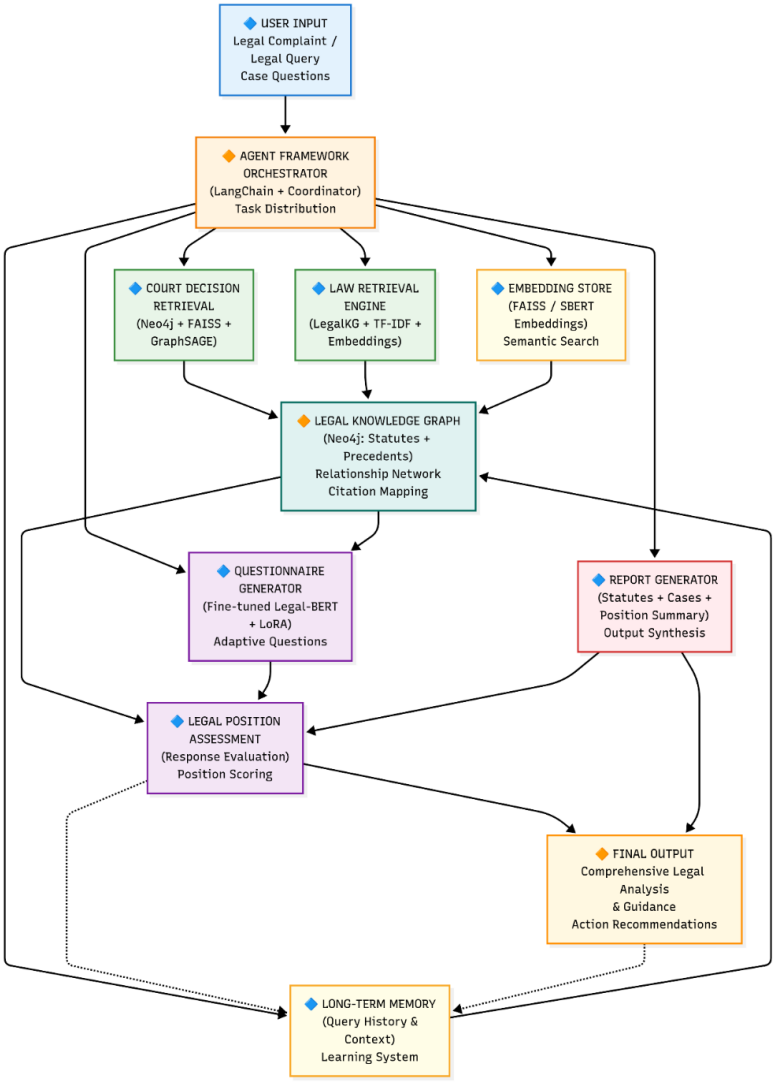

Figure 1.3.1 System Overview Diagram

MY COMPONENT SYSTEM DIAGRAM

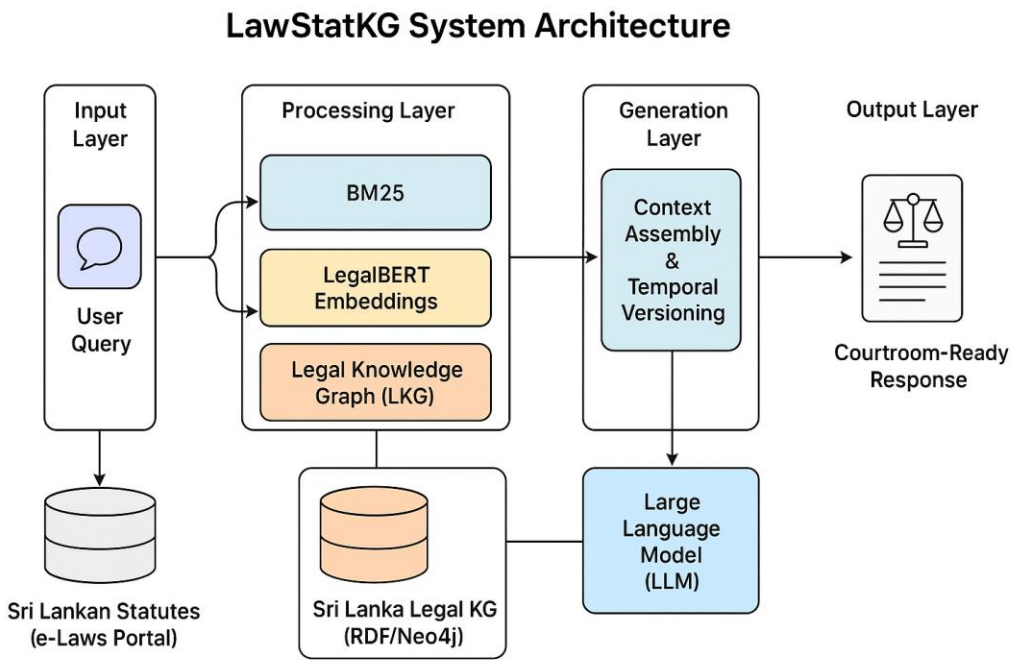

Figure 2.3.2 Component system diagram

LawStatKG — System Architecture Overview

1. Input Layer

NLL Query (laws that govern illegal land occupation)

Filter Options (jurisdiction, date range)

2. Data Sources Layer

Statute Corpus (text-based form of the laws)

Gazette Amendments Database

Related Acts Repository

Metadata Store (jurisdictions, dates, categories)

3. Processing Layer

Query Preprocessing

Hybrid Retrieval Engine

BM25 Lexical Search (match retrieval)

LegalBERT Semantic Search (meaning-based retrieval)

Knowledge Graph Access Layer

RDF/Neo4j Database Query

Entity & Relationship Retrieval

Temporal Tracking Module (history of versions, dates of enactment)

4. Output Layer

Consolidated Statute View (incorporated with amendments)

Relationship Map Visualization (graphical output of legislations with relationship)

Temporal Timeline View (historical evolution of the law)

Courtroom-Ready Report (PDF/HTML output).

3.5 Project Flow

1. Data Acquisition

Firstly, collect authoritative evidence of enactment, Major enactments of the e-Laws web site of the Sri Lanka Parliament. Gazette notices including amendment, repeal, and subsidiary legislation are noticed. Law reports and PDF archives available to the general public were connected to legal sources. PDF-to-text and web scraping will be employed if necessary to form a homogeneous corpus of law. All of the sources will be stored permanently in a metadata repository with publication date, legal topic, and gazette number so that temporality can be easily tracked when it must be done in the future.

2. Preprocessing and Normalization

Preprocessing cannot be avoided because of heterogeneity of Sri Lankan legal data. Steps are: Optical Character Recognition (OCR) of scanned gazette pages. Text normalization and cleaning (removing formatting artifacts, OCR correction, and terminological standardization.) Segmentation of the laws into clauses, subsections, and sections. Metadata tagging, enactment date, amendment date, and related laws, using rule-based and machine learning algorithms.

3. Development of Hybrid Retrieval Engine

The retrieval sub-system will have: Hybrid search engine: Lexical search through BM25 for exact keyword matches. Semantic search through a Sri Lankan legal text trained LegalBERT model for contextual sense. Query interpretation to execute natural language

queries and map them onto relevant legal provisions. Ranking fusion to combine lexical and semantic scores to get best retrieval precision.

4. Legal Knowledge Graph Construction

We will create a domain-specific RDF-based knowledge graph to model legal relations. Ontology building to define entities (statute, amendment, provision) and relations (has amendment, refers to, related to). Entity linking to link statutes to their amendments, related acts, and cross-references. Temporal versioning to store historic amendments, hence being capable of querying the laws on a specific date. Graph database deployment via Neo4j or an RDF with SPARQL query support.

5. Consolidation and Report Generation

The software must have an auto-document composition facility that: Combines the initial legislation with all the appropriate amendments. Places the legislation as it stands currently, its previous changes at will are available. Generates court-proof PDF or HTML output with references, cross-references, and context.

6. Testing

This process is iterative. Feedback obtained through the user testing in the last step may initiate changes in the earlier steps, i.e., adjusting the knowledge graph scheme or retraining LegalBERT model.

Testing comprises:

Unit Testing: Component-level testing

Integration Testing: Entire system is tested

Real-World Testing: Operating in law environment.

User Testing: Lawyers' usability and feedback

3.6 Commercialization

The advice would have been that such a system be launched as a stand-alone entity, independent of a cooperative organization, mainly because there is no centralized efficient platform in Sri Lanka that connects legal professionals to AI tools. Any lawyer, regardless of their firm or affiliation, would be free to use the system this way. Multi-device system design will be the next step with a mobile app for lawyers to access the system's features. Probably, this product will follow a freemium business model. The free version could have

the basic functionality of generating questions or set a limit on the number of questions per case. The premium version would come with greatly enhanced features such as:

• Unlimited and elaborate question generation that deepens the legal context and nuance.

• This connectivity feature expands the legal knowledge base along with real-time legal updates.

• The advanced fact elicitation and iterative refinement provide the user with additional assistance in research for the complimentary legal practice.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

1. Keyword (BM25) and semantic (LegalBERT) search of statutes.
2. Knowledge graph of statute relationship (amendments, references) to store.
3. Show historical versions of statutes to monitor.
4. Show relationship-aware results with graph visualization.
5. Generate one, courtroom-ready report with statute text, amendments, and relationships.
6. Export as PDF or Markdown.

4.2 Non-functional Requirements

1. Performance – Show results in less than 3 seconds.
2. Scalability – Handle larger statutes efficiently.
3. Security – Use web based and role-based access control.
4. Reliability – Give 99% uptime supported with backups.
5. Maintainability – Documentation and Modular development
6. Integration – Used REST APIs.
7. Concurrency – Handle 500+ concurrent users.

4.3 User Requirements

1. Reduce statute research time from hours to minute.
2. Provide up-to-date laws with clear amendment history.
3. Claims to comprehend legal principles, not keywords alone.
4. Demonstrates how laws relate to each other in an easy-to-understand visual way.
5. Generates professional, ready-to-use output for court.

5. BUDGET DETAILS

Description	Cost	Occurring
Server Hosting	Covered BY GitHub Pack	Monthly
Cloud Database	Covered BY GitHub Pack	Monthly
Legal Metirials	LKR 10,000	Annual
Developer Tools/Software Subscriptions	Covered BY GitHub Pack	Monthly/Annual
LLM API usage	LKR 10,000	Monthly

Table 2.1 Budget for the proposed system

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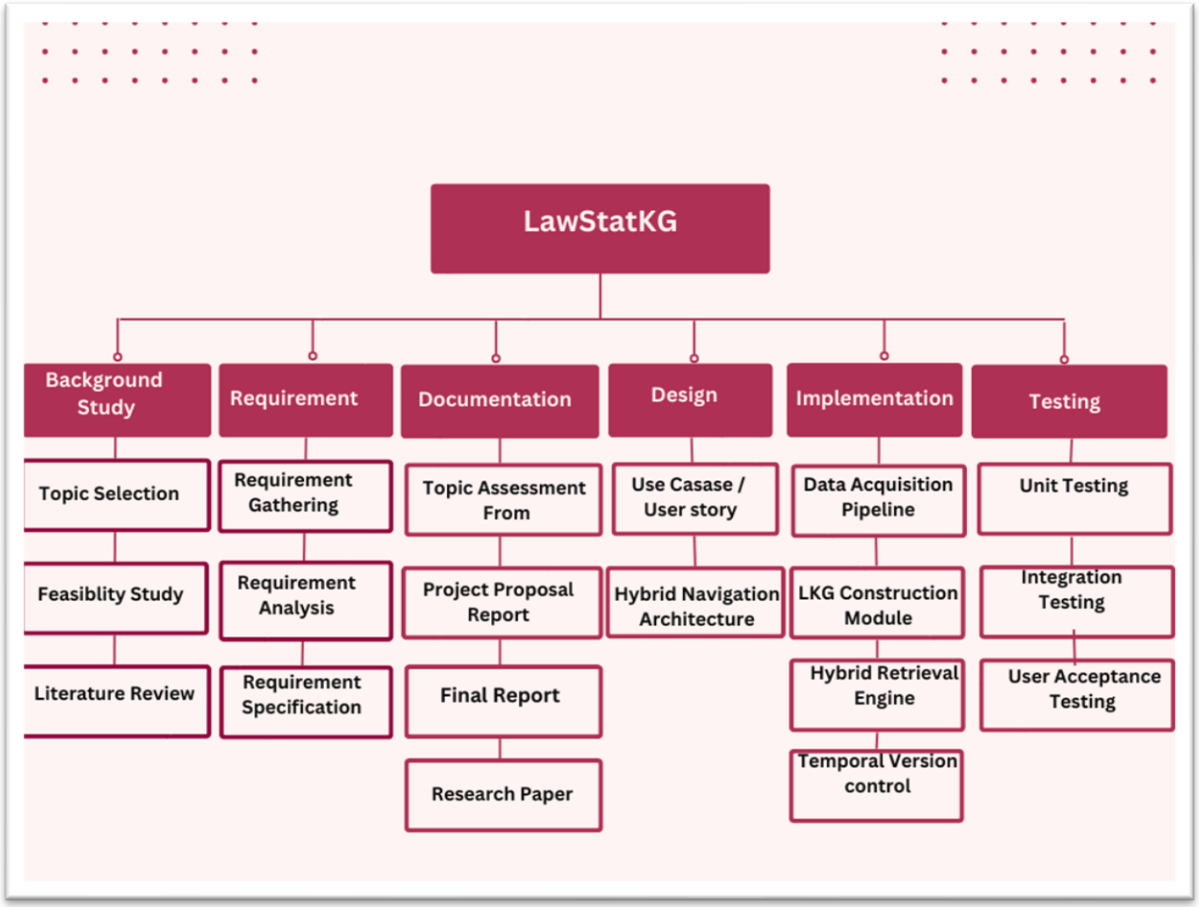
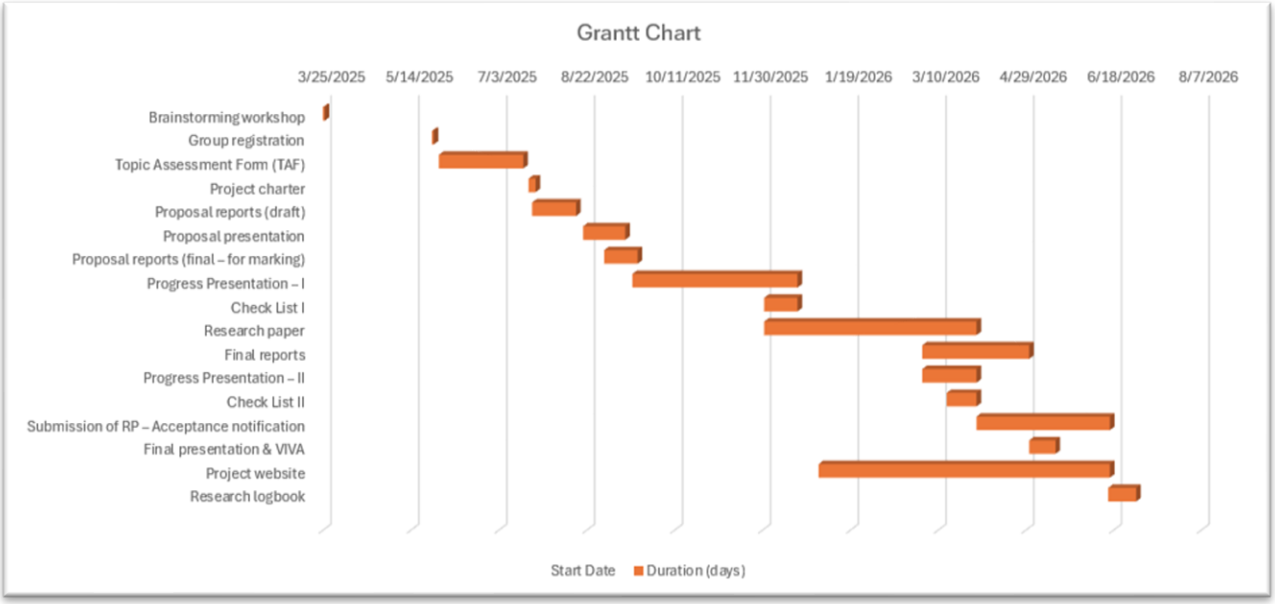
APPENDICES

Appendix 1. Work Breakdown Structure (WBS)



Appendix 2. Gantt Chart



Appendix 3. Similarity Report

Proposal draft

ORIGINALITY REPORT

5%	4%	1%	4%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to Sri Lanka Institute of Information Technology Student Paper	2%
2	louisdl.louislibraries.org Internet Source	1%
3	Submitted to The Robert Gordon University Student Paper	<1%
4	Submitted to UCL Student Paper	<1%
5	Submitted to The Shah Noorani Institute of Technology (SNIT) Student Paper	<1%
6	www.tdx.cat Internet Source	<1%
7	www.coursehero.com Internet Source	<1%

LEGAL RESEARCH AND EVIDENCE ASSISTANT – JURIAID

Project ID: 25-26J-274

PROJECT PROPOSAL REPORT

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BSc. (Hons) Degree in Information Technology

Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

July 2025

LEGAL RESEARCH AND EVIDENCE
ASSISTANT – JURIAID

Project ID: 25-26J-274

PROJECT PROPOSAL REPORT

BSc. (Hons) Degree in Information Technology Specializing in Information Technology

Department of Information Technology

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

Name	Student ID	Signature
Sammani W.M.W.L.	IT22318848	

Statement of the supervisor,

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor:

Date:
29/08/2025

Signature of the co-supervisor:

Date:
29/08/2025

Abstract

This project addresses a significant challenge faced by legal practitioners: the time-consuming and manual process of creating legally relevant questions (for acceptance and resolutions) during case management. Currently, there is no dedicated platform to automate this crucial step, leading to inefficiencies and increased workload for lawyers.

Our proposed solution is an AI-powered legal assistant system designed to generate these questions automatically. The system will operate by taking an initial case input, such as a complaint and the victim's reply, and then leveraging legal-specific knowledge bases (past cases, statutes, and laws) to formulate precise and contextually appropriate follow-up questions.

The core design involves fine-tuned legal language models (e.g., Legal-BERT variants), Retrieval-Augmented Generation (RAG), and context-aware fact elicitation. This will enable the system to identify missing legal facts, generate dynamic clarifying questions, and produce legally formatted outputs grounded in authoritative sources.

By automating this process, the research aims to significantly enhance the efficiency and accuracy of legal case management, reduce the manual effort involved, and provide lawyers with a powerful tool for delivering more personalized and effective legal services. The system will prioritize reliability, security, usability, and scalability to ensure practical applicability within the legal domain.

Keywords: Legal AI, Question Generation, Retrieval-Augmented Generation, Legal- BERT.

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1. INTRODUCTION

In modern legal practice, managing case files efficiently is paramount. The initial stages of a legal case involve a client providing a complaint to their lawyer, who opens a case file. A copy is sent to the defendant, whose lawyer provides a reply. A critical subsequent step requires lawyers to manually create a series of "acceptance" and "resolution" questions to gather further details, clarify ambiguities, and strategize.

However, this manual process is highly time-consuming and lacks a dedicated, automated platform [1]. Lawyers spend considerable effort drafting these questions, effort that could be allocated to complex legal analysis. This manual effort also introduces variability in the completeness and precision of the questions, potentially prolonging the legal process.

This research project proposes to develop JuriAid, an AI-powered system to address this critical need by automating the generation of these legal questions. The system will analyze the initial complaint and reply, and by leveraging past cases and relevant laws, it will intelligently formulate comprehensive and context- aware questions. Our aim is to significantly uplift the efficiency and effectiveness of legal case management.

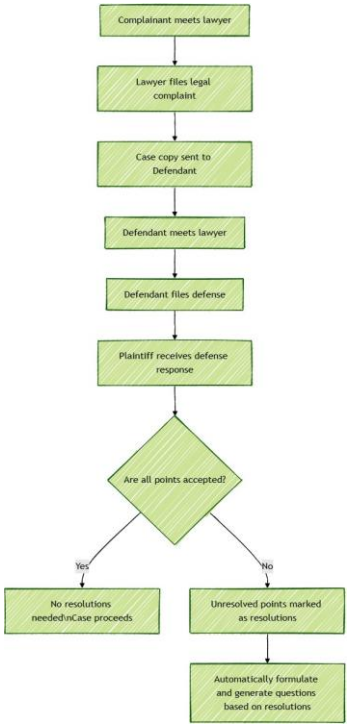

Figure 1.1 flowchart of the scenario

2. LITERATURE REVIEW

The legal domain is increasingly adopting Information and Communication Technologies (ICTs) to enhance efficiency and access to justice [1]. Traditional legal workflows, heavily reliant on manual research and document drafting, are labor-intensive and time-consuming, which highlights the necessity for AI- assisted solutions. Domain-specific models such as Legal-BERT [2] have demonstrated superior performance over generic BERT models by capturing the nuances of legal vocabulary and structure. While Legal- BERT excels in comprehension and classification tasks, it is limited in its ability to generate dynamic, context-aware questions for initial case management, illustrating a gap between text understanding and interactive fact elicitation.

To address issues of factual accuracy and grounding, Retrieval-Augmented Generation (RAG) [3] [4] architectures have been employed, effectively combining dense retrieval with generative models to produce outputs grounded in verified knowledge bases. The implementation of RAG in systems like SCaLe-QA [3] in Sri Lanka demonstrates that domain-tuned embeddings and semantic chunking can enhance retrieval performance in a localized legal context. Although these systems strengthen factual reliability, they focus on question answering rather than the iterative, client-interactive process necessary for case intake, highlighting a functional limitation when compared to the needs of dynamic question generation.

Complementing comprehension and grounding, research on interactive fact elicitation and automated drafting provides insight into simulating aspects of lawyer-client interactions. For example, Lin and Cheng (2024) [5] illustrate that fine-tuned language models can produce structured judicial documents while generating clarifying questions based on incomplete contexts. This capability suggests that AI can partially replicate human follow-up questioning, bridging the gap between static document generation and dynamic fact-gathering. However, existing models prioritize formal document output rather than the iterative elicitation process, revealing an unaddressed need for a system that integrates comprehension, factual grounding, and interactive questioning in a single workflow.

Examining these approaches comparatively, it becomes evident that each address only a part of the broader challenge. Legal-BERT [2] ensures domain-specific understanding, RAG guarantees factual grounding, and fact-elicitation models enable question generation [5].

Individually, they provide robust functionality; collectively, however, no existing system integrates these capabilities into a unified tool for automating initial case intake within the Sri Lankan legal context. Systems like SCaLe-QA [3] excel at question answering but lack generation capabilities, whereas drafting-focused approaches [5] automate document creation but do not support interactive questioning. This cross-analysis underscores the novelty and relevance of JuriAid, which aims to synthesize these proven technologies into an integrated solution that addresses a clearly defined workflow gap, enhancing efficiency, accuracy, and comprehensiveness in early-stage legal procedures.

Table 2.1 table of analysis

Aspect	Traditional Legal Processes	Existing AI Approaches	Strengths	Limitations / Gaps	Implications for JuriAid
Research & Document Drafting	Manual review of case law, statutes, and prior judgments; labor-intensive and time-consuming.	Legal-BERT [3]enables domain-specific understanding through pre-training on legal corpora.	Captures nuanced legal vocabulary; superior performance on legal downstream tasks.	Focused primarily on comprehension and classification; does not automate question generation.	Provides the semantic foundation for JuriAid to interpret legal context accurately.
Information Retrieval & Knowledge Grounding	Lawyers manually consult multiple sources to verify facts.	Retrieval-Augmented Generation (RAG) [3] [4]; systems like SCaLe-QA (Sri Lanka) leverage semantic embeddings and chunking of case law.	Ensures generated content is factually grounded; mitigates hallucinations.	Primarily applied to question answering, not interactive or iterative question generation.	RAG forms the backbone of JuriAid's knowledge grounding, enabling accurate, verifiable questions.
Automated Question & Fact Elicitation	Lawyers ask follow-up questions to fill gaps in case details.	Interactive fact elicitation models [5] and format-aware drafting (Lin & Cheng, 2024) enable generation of clarifying questions or structured documents.	Models can simulate lawyer-client interactions and draft legal outputs automatically.	Focused on structured document drafting rather than dynamic, iterative interview-style question generation.	JuriAid integrates this approach to produce dynamic, context-aware initial case management questions.
Integration of Technologies	Highly fragmented; humans coordinate comprehension,	Individual models (Legal-BERT, RAG,	Each component is highly	No single system integrates understanding, grounding, and	JuriAid synthesizes these technologies into a

	retrieval, and questioning.	fact-elicitation LLMs) exist.	effective in isolation.	question generation in a Sri Lankan context.	cohesive system tailored for automated case question generation, addressing a specific workflow gap.
Research Context	Global legal AI research often focuses on English-speaking or high-resource jurisdictions.	Some local adaptation exists (SCaLe-QA in Sri Lanka) but limited to QA tasks.	Localized models demonstrate the feasibility of adapting AI to specific legal systems.	Lack of integrated, domain-specific solutions for initial case intake and dynamic question generation.	Highlights the novelty and practical relevance of JuriAid in addressing an unmet need in Sri Lanka's legal system.

3. RESEARCH GAPS

A research gap is an area within a field of study that has not been thoroughly investigated or addressed in previous research, representing a missing element in the existing knowledge [1].

The major research gap identified in current legal practice is the absence of a dedicated and intelligent automated platform for generating legally relevant questions (acceptance and resolutions) based on case file details and legal precedents [1]. Lawyers currently rely on a manual and time-consuming process to formulate these questions, a problem explicitly stated by legal professionals.

Existing legal tech solutions may offer general legal research tools or document automation for standard templates, but they often lack the sophisticated, context-aware intelligence required to dynamically generate specific, targeted questions for individual case files by integrating historical case data and statutes in an integrated manner [3] [1]. This leaves a significant void in automating a critical, repetitive, and expertise- driven task in the legal workflow. The current reliance on manual question creation means there's no systematic way to ensure comprehensive coverage, consistency across cases, or rapid adaptation to new legal nuances without substantial human effort. This gap prevents lawyers from fully leveraging technology to enhance efficiency in the initial and ongoing stages of case assessment [1].

4. RESEARCH PROBLEMS

Lawyers currently spend significant time manually generating legally relevant questions during case preparation, a process that is both inefficient and prone to human error. The absence of an automated mechanism increases workload and delays case management. Hence, there is a critical need for an AI-based system capable of automatically generating contextually accurate and legally valid questions to improve efficiency, consistency, and overall quality in legal practice.

Based on the identified research gap, the core problems this project seeks to resolve are:

• **How to accurately and efficiently generate legally relevant questions (acceptance and resolutions) for case files?**

The current manual process is time-consuming and prone to variations in completeness [1]. The challenge lies in developing an AI system that can ingest diverse legal case details (e.g., complaint, victim's reply) and then intelligently formulate precise, structured questions that cover all necessary legal facts for clarity and resolution

• **How to ensure the generated questions are contextually appropriate, legally sound, and comprehensive, mimicking a lawyer's expertise?**

Generic AI models often struggle with the nuances of legal language and context [2]. The problem involves designing the system to integrate domain-specific knowledge, adhere to legal standards (e.g., IRAC structure implicitly), and anticipate missing information to generate follow-up questions effectively.

• **How to effectively leverage vast amounts of past case data and legal statutes to inform the question generation process?**

Lawyers currently rely on their experience and manual research of precedents. The challenge is to create a robust retrieval and reasoning mechanism that can quickly access and apply relevant legal authority and historical case patterns to the automated question generation, ensuring questions are grounded in established legal principles and precedents [3] [6]

5. OBJECTIVES

Main Objectives

The main objective of this research is to **develop an AI-powered legal assistant system** that automates and enhances the generation of context-aware, legally sound questions (acceptance and resolutions) for effective legal case management, thereby **significantly improving the efficiency and accuracy** for legal practitioners in Sri Lanka and potentially beyond. This innovation intends to better support lawyers in streamlining their workflow, reducing manual effort, and ensuring comprehensive information gathering throughout the case process.

Sub Objectives

To achieve the main objective, the following specific sub-objectives must be accomplished:

• **To design and implement a fact-elicitation module** capable of analyzing initial case inputs (complaints, replies) to identify and automatically elicit missing key legal facts (e.g., dates, parties, clauses) required for comprehensive question generation. [5]

• **To develop a grounded legal question generation engine** that leverages retrieval-augmented large language models (LLMs) to produce precise questions backed by relevant statutes, case law, and precedents, thereby reducing inaccuracies and improving reliability. [2] [4] [3]

• **To create an intelligent document retrieval system** capable of fetching relevant statutes, precedent snippets, and legal documents via semantic retrieval systems (e.g., Legal-BERT embeddings) and lexical methods (e.g., BM25) to inform the question generation process. [3] [7]

• **To provide a user-friendly interface for lawyers** to input case details, review generated questions, and interact with the system for clarifications and refinements, ensuring seamless integration into existing legal workflows.

6. METHODOLOGY

The development of this AI-powered legal question generation system will follow a systematic methodology, encompassing data collection, model development, system integration, and rigorous testing. The overall system aims to provide a comprehensive solution for legal question automation.

Overall System

The whole system, as described by your context, can be conceptualized as an AI-powered legal assistant with a core component for question generation. Referring to the provided system diagram, the overall architecture of such a legal AI system would include: [3] [7]

• User Input: Lawyers provide initial case details, complaints, and replies.

• Agent Framework Orchestrator: Manages task distribution and overall workflow.

• Court Decision Retrieval & Law Retrieval Engine: These components are responsible for fetching relevant statutes, precedents, and case law from a comprehensive legal knowledge graph (e.g., Neo4j) and embedding stores (using FATSS/SBERT and embeddings).

• Questionnaire Generator (Core Component): This module, powered by fine-tuned Legal-BERT with LoRA adaptation, dynamically generates context-aware questions and assesses legal positions.

• Legal Position Evaluation & Report Generator: Assesses the legal position based on input and generated questions, leading to output synthesis (e.g., draft questions, summaries).

• Long-Term Memory: Stores query history and aids in continuous learning.

• Final Output: Comprehensive legal analysis, guidance, and actionable recommendations (in your case, the generated legal questions).

Individual Component Diagram (Legal Question & Q&A Component)

The core "Legal Questionnaire & Q&A" component, which focuses on generating questions, operates through a pipeline: [3] [5]

1. Input: User (lawyer) submits initial case details (e.g., a complaint, reply).

2. Retrieval: The system uses Legal-BERT embeddings to query case-law and statute indexes, retrieving relevant chunks of legal information. This is done via a domain-tuned retrieval

system.

3. Analysis: A fine-tuned LLM (e.g., Legal-BERT-based) analyzes the user input along with the retrieved legal information. If crucial facts are missing, it generates targeted, clarifying questions for fact elicitation.

4. Follow-Up: The user provides answers to the generated questions, and the LLM may iterate by asking further clarifying questions until sufficient context is gathered.

5. Generation: Once enough context is available, the LLM generates the final output, which in your system would be the legally formatted "acceptance and resolution" questions, citing relevant authorities.

6. Feedback: The system may prompt for review or flag areas needing validation from the lawyer.

Key Tasks Involved:

• Fact Elicitation: Assessing user input for missing key legal facts and auto-generating focused follow-up questions. [5]

• Question Generation: Producing structured, legally relevant questions (e.g., "Date of incident?", "Jurisdiction?").

• Document Retrieval: Fetching statutes and precedent snippets via semantic retrieval systems (LEGAL- BERT embeddings) and lexical BM25 methods.

• Answer Generation / Drafting: Using retrieved documents as prompt context for generating grounded responses or draft documents, maintaining legal formatting (e.g., the final questions for lawyers).

• Feedback & Refinement: Iterative cycles where the LLM may ask for clarifications or regenerate text based on new inputs or system checks.

Technology Stack: The project will utilize Legal-BERT variants and other specialized LLMs fine-tuned on legal corpora, Retrieval-Augmented Generation (RAG) frameworks, potentially knowledge graphs (e.g., Neo4j) for structured legal information, and advanced NLP techniques for text processing and semantic understanding. The system will also handle multilingual inputs, specifically Sinhala and English, as relevant to the Sri Lankan legal context. The user interface for lawyers would likely be a web application (e.g., using Next.js for therapists in other similar projects).

Figure 6.1 system diagram

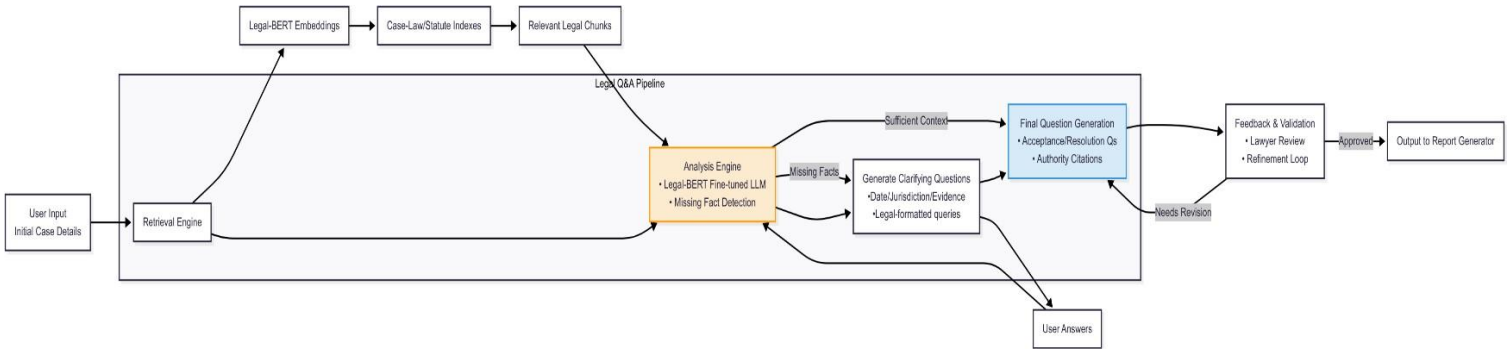

Figure 6.2 component diagram

7. PROJECT REQUIREMENTS

Functional Requirements

Functional requirements outline the precise actions, duties, or responsibilities that the system must carry out. For the legal question generation component, these include:

• Case Input Processing: The system shall accept diverse inputs related to a legal case, including initial complaints, victim replies, and other relevant case details from the lawyer.

• Fact Elicitation: The system shall analyze the provided case input to identify and highlight missing crucial legal facts (e.g., names, dates, locations, specific clauses). [5]

• Dynamic Question Generation: The system shall automatically generate a series of legally relevant and context-aware questions (acceptance and resolutions) based on the processed case details and identified missing facts.

• Legal Knowledge Retrieval: The system shall retrieve relevant legal statutes, precedents, and past case decisions from a comprehensive legal knowledge base to ground the generated questions in authoritative sources. [3] [6]

• Contextual Adaptation: The system shall adapt the generated questions based on iterative feedback from the lawyer, allowing for refinement and deeper inquiry into specific aspects of the case.

• Output Formatting: The system shall present the generated questions in a clear, structured, and legally appropriate format, suitable for direct use by lawyers. [5]

• Multilingual Support: The system should support processing inputs and generating questions in multiple languages relevant to the legal context, such as Sinhala and English.

Non-Functional Requirements

Non-functional requirements define the criteria that describe the system's behavior and quality attributes.

• Security: The system must implement robust security measures to protect sensitive client information and confidential legal documents from unauthorized access, breaches, and threats.

• Reliability: The system must be highly reliable, ensuring consistent availability for lawyers to use at any time, with minimal downtime or errors during question generation.

• Efficiency: The system must significantly reduce the time lawyers spend on manually generating questions, ensuring quick processing of inputs and rapid generation of questions. [2]

• Usability: The system's interface must be intuitive and user-friendly for legal practitioners, allowing for easy input of case details and straightforward review of generated questions.

• Scalability: The system must be designed to accommodate a growing number of users (lawyers) and an increasing volume of case data without compromising performance or response times.

• Accuracy: The generated questions must be legally accurate, relevant, and comprehensive, reflecting the nuances of legal reasoning and complying with legal standards. [2] [3]

• Explainability: Where possible, the system should provide insights into the reasoning behind generated questions or highlight the legal sources that informed its output, enhancing user trust (implicitly covered by "grounded generation" and "citation-aware" features).

8. COMMERCIALIZATION

Budget

The following amounts are estimated and may be subject to change:

Table 8.1 budget table

Description	Cost	Occurring
Server Hosting	Covered by GitHub Pack	Monthly
Cloud Database (e.g., AWS RDS for legal data storage)	Covered by GitHub Pack	Monthly
Legal Materials	LKR 10,000.00	Annual
Developer Tools/Software Subscriptions	Covered by GitHub Pack	Monthly/Annual
LLM API Usage	LKR 10,000.00	Monthly

Commercialization Methods

It is advised to launch this system as a **stand-alone entity** rather than being bound to a cooperative organization, especially given the current lack of a centralized, efficient platform in Sri Lanka that connects legal professionals with such AI tools. This approach ensures that any lawyer, regardless of their firm or affiliation, can utilize the system.

The system will be designed for **multi-device support**, with a dedicated **mobile application for lawyers** to access its features. It will likely adopt a **freemium business model**. A free tier could offer basic question generation functionalities or a limited number of questions per case. The premium tier would unlock advanced features crucial for comprehensive legal practice, such as:

- **Unlimited, sophisticated question generation** with deeper legal context and nuance.
- **Integration with broader legal knowledge bases** and real-time legal updates.
- **Advanced fact elicitation and iterative refinement capabilities.**

9. GRANTT CHART

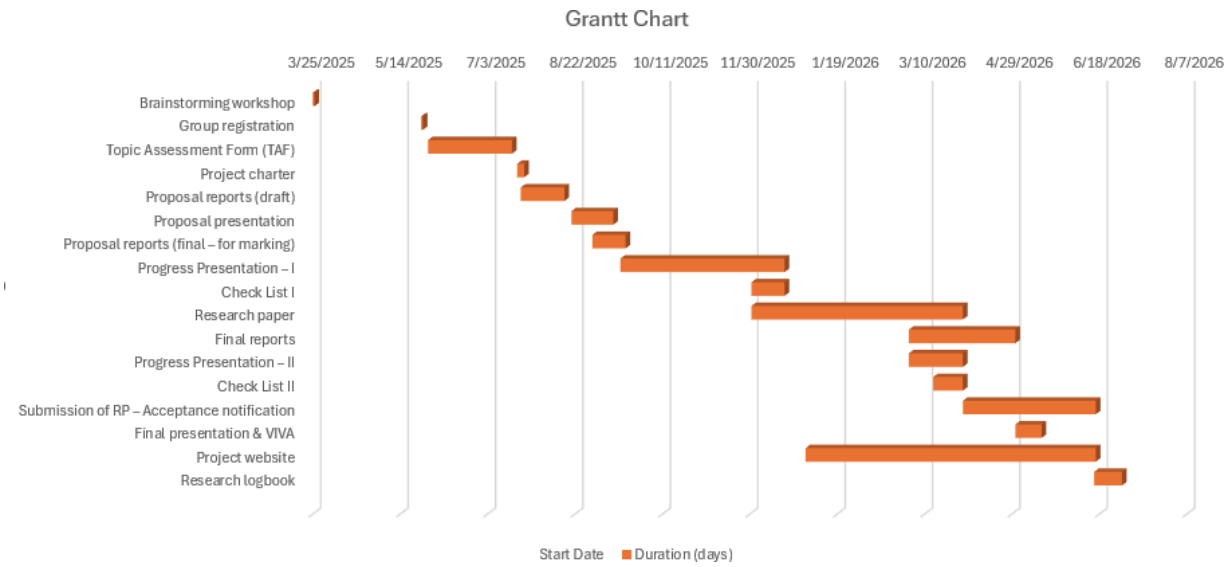

Figure 9.1 Grantt chart

10. Work Breakdown Structure

Figure 10.1 WBS chart

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12.Appendices

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LEGAL RESEARCH AND EVIDENCE ASSISTANT - JURIAID

Project ID: 25-26J-274

PROJECT PROPOSAL REPORT

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BSc. (Hons) Degree in Information Technology

Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

July 2025

JuriAid: An AI-Powered Legal Assistant System for Automated Question Generation in Legal Practice

BSc. (Hons) Degree in Information Technology Specializing in

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

Name	Student ID	Signature
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Statement of the supervisor,

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor:

Date: 29-08-2025.

Signature of the co-supervisor:

Date: 29/08/2025

Abstract

Legal help in Sri Lanka isn't available where people need it most. You'll find experienced lawyers concentrated in Colombo and a few other major cities, but if you live in a rural area, good luck finding quality legal assistance. Meanwhile, the lawyers who do practice still waste countless hours on manual research - flipping through books, searching physical files, doing things the same slow way they did decades ago. Here's what we noticed: nobody has built a smart system that can actually handle the messy, complicated work that legal cases require while managing different AI tools designed specifically for how Sri Lankan law works. So, we decided to build one ourselves. We're using something called LangChain to create what's basically a digital legal coordinator. Picture this: someone comes to you with a legal problem. Instead of spending hours doing everything yourself, this system takes that problem, chops it up into pieces that make sense, and hands each piece to a different specialist. One specialist digs through old court cases, another figures out which laws apply, and a third one asks smart questions to fill in the gaps. We built in a memory system so it remembers what happened in previous conversations, plus a thinking component that mimics how seasoned lawyers actually work through problems. When everything runs smoothly together, you get comprehensive legal analysis that actually fits the situation. Why does this matter? We're not just tweaking something that already exists - we're building something brand new that can do the heavy lifting legal experts normally do by hand. Practicing lawyers should be able to work faster and prepare better cases. More importantly, regular people across Sri Lanka might finally be able to get decent legal advice without traveling to the capital.

Keywords: Legal AI, Multi-agent Systems, Legal Accessibility, Sri Lankan Jurisprudence, LangChain Framework

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
LLM	Large Language Models
MoE	Mixture-of-Experts
NLP	Natural Language Processing
RMA	Reflective Multi-Agent
SOP	Standard Operating Procedure

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Appendix A : Turnitin Similarity Score 18

1. Introduction

Sri Lanka's justice system has a serious problem that most people don't talk about enough - there's a huge gap between who needs legal help and who can actually get it. Here's what's happening: more than 70% of all practicing lawyers work in Colombo and a few other big cities [1]. If you live anywhere else in the country, finding a decent lawyer becomes incredibly difficult. And even when lawyers are available, they're still doing research the same way people did it thirty years ago - manually searching through books and basic digital databases that can't really understand what they're looking for [2].

We kept asking ourselves: why hasn't anyone built something better? That's where the 'JuriAid' project comes in. Our approach moves beyond simply scanning documents and calling it "digital transformation." We aim to develop a system capable of reasoning through legal issues in a manner similar to seasoned legal practitioners. This technology is intended not as a replacement for lawyers, but as a supportive tool designed to manage laborious preliminary work, thereby freeing up legal professionals to focus on strategic analysis [3].

However, building a functional AI system for the legal domain requires more than just bundling different AI tools together and hoping they cooperate. A central intelligence must be in place to direct their actions. This is the precise function of our 'Agentic Framework.' This framework essentially acts as the senior partner in a law firm. It's responsible for analyzing a complex case, delegating the various parts to the right specialists, and then weaving their work into a single, cohesive strategy [4].

The success of the whole system hinges on how well this coordination works. It wouldn't matter if we had the most sophisticated individual AI components available; if they cannot cooperate effectively, the system as a whole would be useless. This research project stems from a key insight: getting AI tools to function as a team in the Sri Lankan legal system, with all its unique quirks, demands a smart and capable coordinator. Therefore, our work involves building and testing this LangChain-based Agent**ic AI** Framework to confirm its ability to manage legal AI systems in complex scenarios [5].

1.1. Background & Literature survey

How lawyers use artificial intelligence is going through a massive change. We've moved past simple data storage and search and are now focused on building smarter systems that can act independently. This change comes from new autonomous agents built on Large Language Models (LLMs). The purpose of these agents is to observe their surroundings, think, and then act to meet a goal [6]. What makes these "agentic" systems different from older AI is their ability to plan, recall information, and use tools-skills that are a perfect match for the complex, step-by-step logic of legal work [7].

These agents generally have four key parts: profiling, memory, planning, and action [6]. The Profiling component gives an agent a specific role or "persona" that shapes its behavior, a core idea in legal AI. For instance, systems like Chatlaw and LawLuo mimic a law firm's structure by creating agents for roles like 'Legal Researcher' or 'Senior Lawyer' [8], [9]. The Memory component is essential for keeping track of long and complex legal cases. This includes short-term memory (what the LLM is currently processing) and long-term memory, which is often handled by external tools like Redis [6], [10].

The Planning function is the agent's strategic core, breaking down large tasks into smaller, more manageable steps. The Action function is what allows the agent to interact with its environment, for example, by using an API to retrieve data from a legal database or knowledge graph [11], [9].

A major downside of LLMs is their habit of "hallucinating," generating information that appears credible but is actually false-a problem that multi-agent systems aim to tackle [8], [12]. The danger of these hallucinations became very real when a lawyer submitted a brief with fake cases made up by ChatGPT [13]. For this reason, the work on multi-agent systems is vital; it's less about a tech upgrade and more about building tools people can trust to be safe and more reliable. In response to these issues, researchers have created several key frameworks.

To address these challenges, researchers have developed several notable frameworks. Chatlaw functions as a virtual legal assistant that adheres to a strict set of operating procedures, much like a real law firm. It uses a Mixture-of-Experts (MoE) model and leans on knowledge graphs to make sure the information it provides is accurate [8]. LawLuo also mimics a law firm's structure, but its

focus is on offering personalized legal help over a series of conversations. Its most distinctive feature is a "Boss" agent that reviews the work of other agents, offering a practical way to incorporate feedback into the planning process [9]. PAKTON is a specialized tool designed for analyzing lengthy legal agreements. Its main technical contribution is a more sophisticated document processing method that uses Hierarchical Parsing and Context-Aware Chunking to maintain the logical integrity of a contract during information retrieval [14]. Finally, the Reflective Multi-Agent (RMA) [15] framework offers a different model focused on internal quality assurance. It uses two specialized agents for review: a Factor Analyst to verify facts and an Argument Polisher to refine the quality of the legal arguments.

Looking at these systems, two different approaches become clear. Some, like Chatlaw and LawLuo, try to copy how a law firm works ("Process Emulation"). Others, like PAKTON and the RMA framework, focus on copying how a lawyer thinks ("Cognitive Emulation"), including the process of questioning and self-correction.

1.2. Research Gaps

After diving deep into everything that's been published about LLM-based autonomous agents in legal work, we've spotted two major holes in the research that nobody seems to be talking about much. These gaps are exactly what we're trying to tackle with the JuriAid Agentic AI Framework.

The first problem is technical. All these impressive systems we've been reading about are great at what they do, but they're all missing something crucial. Most of them just follow a basic script: first do this, then do that. ChatLaw, for example, has what they call a Standard Operating Procedure (SOP) that's basically just a flowchart telling each agent when it's their turn to work [8]. But real legal work doesn't follow flowcharts [16]. What we really need is something like a senior partner in a law firm - someone who can look at a legal problem, break it down intelligently, and then coordinate all the junior associates to work on the right things in the right order, changing the plan when new information comes to light. None of the existing systems can do this kind of real-time strategic coordination [6], [7].

The second gap is contextual. Almost everything published in this field focuses on either American/British contract law or Chinese legal systems [7], [8], [9]. But what about everywhere else? Sri Lanka is a perfect example. We have our own statutes, our own court precedents, our own legal procedures. Nobody has seriously tried to build advanced AI legal systems for contexts like ours. The technical challenges are completely different when you're working with limited data and a unique legal system [17]. But the potential impact is huge because these are exactly the places where AI could make the biggest difference in people's lives.

So we're basically trying to solve two problems at once: building smarter agent coordination technology, and proving that this technology can work in legal systems that have been completely ignored by the research community so far.

Table 1 Comparative analysis highlighting the novelty of the proposed system.

Framework	Architecture Type	Key Components	Memory System	Strengths	Limitations	Sri Lankan Law Compatibility
ChatLaw [8]	Process Emulation (MoE)	Legal Researcher, Senior Lawyer agents	Knowledge Graph + Short-term	Structured SOP workflow, Fact verification	Static workflow, Limited adaptability	Low - US/Chinese law focused
LawLuo [9]	Hierarchical Multi-agent	Boss Agent, Specialist agents	Conversation buffer	Personalized consultations, Feedback loops	Linear processing, No long-term memory	Medium - Adaptable structure
PAKTON [14]	Document-centric	Hierarchical Parser, Context Analyzer	Document-based chunking	Advanced contract analysis	Limited to document analysis	Medium - Document processing applicable
RMA Framework [15]	Cognitive Emulation	Factor Analyst, Argument Polisher	Internal review system	Quality assurance, Self-correction	Narrow scope, No case continuity	Low - Specialized for arguments only
JuriAid (Proposed)	Dynamic Orchestration	Agentic AI Framework, Memory Manager, Reasoning Engine	Redis + ConversationBuffer	Adaptive planning, Sri Lankan law focus, Long-term memory	Under development	High - Designed specifically for Sri Lankan jurisprudence

1.3. Research Problems

Our review of the current research on multi-agent systems for legal AI has led us to a core question that remains largely unanswered:

How can we build a central coordination system that manages multiple specialized AI agents for legal work, keeps track of complex conversations, and mimics how experienced Sri Lankan lawyer's approach legal problems?

This question is the foundation of our research. We aim to do more than just connect different AI tools; we want to build a system that can manage a legal consultation from the initial client meeting to the final advice. Our research is designed to tackle four specific technical hurdles:

Smart Task Management: Most legal AI systems currently work in a very linear fashion, a direct result of their static workflow designs [8]. Our system needs to make strategic decisions and be flexible enough to change course when the initial analysis uncovers something unexpected.

Conversation History Management: Legal consultations are like extended detective work. Most AI systems treat each question as if it exists in a vacuum, a known challenge in autonomous agents [6]. We need something that can follow the thread of a long, complicated legal conversation.

Replicating Legal Thinking: How do you teach a computer to think like a lawyer with twenty years of experience? Human lawyers develop this kind of strategic thinking through years of practice [16]. We need to figure out how to build that same adaptability into our AI system.

Adapting to Sri Lankan Law: Almost all current legal AI has been developed for the American or Chinese legal systems [7], [9]. An AI trained for a law firm in New York simply won't work in a courtroom in Colombo. A key part of our research, therefore, is to find ways to adapt these technologies for a legal system that has been completely ignored by the major players in the field.

Solving these problems would result in a system with the potential to transform legal services in Sri Lanka and offer a valuable contribution to the broader field of legal AI.

2. Objectives

2.1. Main Objectives

This research is all about building and testing the "Agentic AI Framework." We want it to act like a smart manager for JuriAid's AI tools, thinking through problems the way an expert lawyer would [16]. The end goal is a tool that gives reliable legal advice that gets the full picture of a case under Sri Lankan law.

2.2. Specific Objective

- **Create a central coordinating system with LangChain.** This will be the project's engine, taking a user's query, breaking it into logical tasks, and sending each task to the right specialist agent (like the Court Retriever, Law Retriever, or Questionnaire Generator) [5].

- **Implement a memory system for ongoing conversations.** To keep a long-term record of each case, we'll use a permanent data store like Redis [11]. This will allow the system to follow extended discussions and manage complex cases that evolve over time without losing important details [6], [10].

- **Create a reasoning engine using a mix of rules and model-based logic.** That engine will be the system's "mind." It will analyze the user's question and the facts of the case to select the right agent for the task, giving the framework its intelligence [6].

- **Use FastAPI to create a reliable API for communication between all system parts.** This will be done by creating a standard set of REST APIs to connect the main coordinator with the specialist agents, which will make the system modular and easy to build on in the future.

- **we will establish a clear plan for handling errors.** Our plan will involve using retries, timeouts, and backup strategies to keep the system running without any interruption, even when things go wrong.

3. Methodology

3.1. Overall System

We're going to build the Agentic AI Framework using a step-by-step process that combines well-known ideas from autonomous agent design [6], [7] with modern software development methods. To really get what the Agentic AI Framework does, it helps to see where it fits into the bigger JuriAid system. It's essentially the project manager for a team of specialized AI tools, a concept explored in multi-agent conversation frameworks [4].

Here’s how it works: When a user asks a legal question, the Agentic AI Framework Orchestrator takes charge. It then uses three other specialist agents to get the job done: one that finds past court decisions, one that looks up laws, and one that generates clarifying questions to ask the user. This multi-agent workflow is similar to architectures seen in other legal AI systems [8], [9].

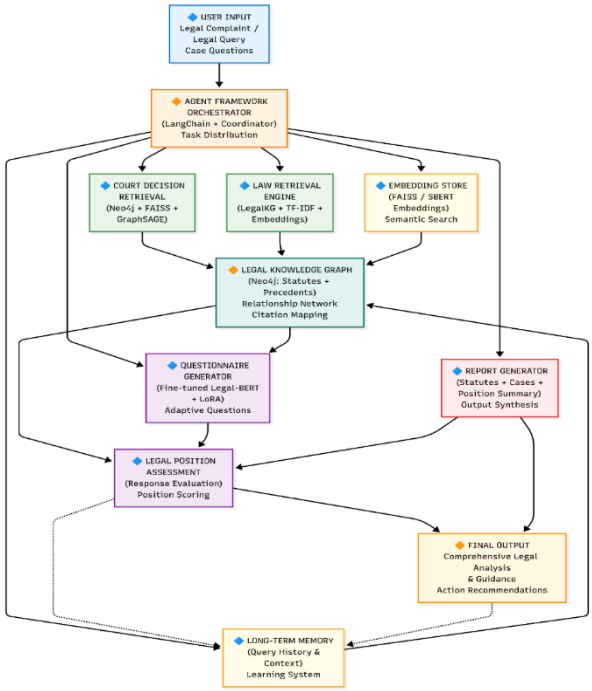

Figure 1:System Overview Diagram

3.2. Individual Component Diagram (Agentic AI Framework)

The core of this research project is the detailed design and implementation of the Agentic AI Framework itself, structured around the unified framework for LLM-based autonomous agents [6]. The implementation will directly map the theoretical modules of an autonomous agent to concrete technologies. The "profiling" of agents is achieved through their explicit, specialized roles. The "planning" module, the central intelligence, will be implemented using LangChain's agent constructs to generate a dynamic, multi-step plan for each user query [5]. The "memory" module will be a hybrid system, using LangChain's “ConversationBufferWindowMemory” for short-term context and a Redis-backed “ConversationBufferMemory” for long-term case continuity [10], [11]. The "action" module is the framework's ability to execute its plan by using the specialist AI agents as "tools," interacting with them through an API management layer [6].

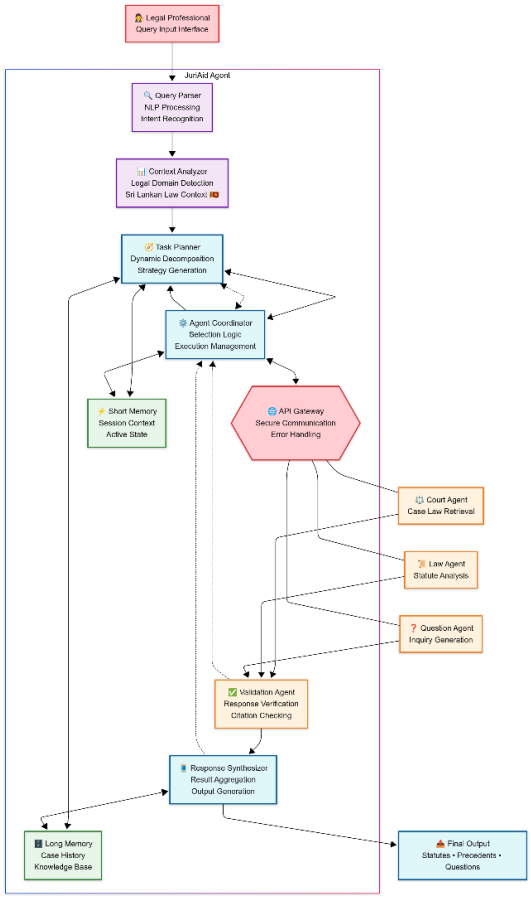

Figure 2 : Component system diagram

3.3. Technology Stack

We've chosen our technology based on tools that are well-established and trusted in the field. The agent's core thinking will be built with LangChain [5] and LlamaIndex [18]. We'll use FastAPI for the API, Redis [11] for long-term memory, and Python for all the coding. This tech combination is designed to handle the entire workflow of a legal question—from understanding the initial query to finding the right documents and asking follow-up questions—all directed by the central orchestrator.

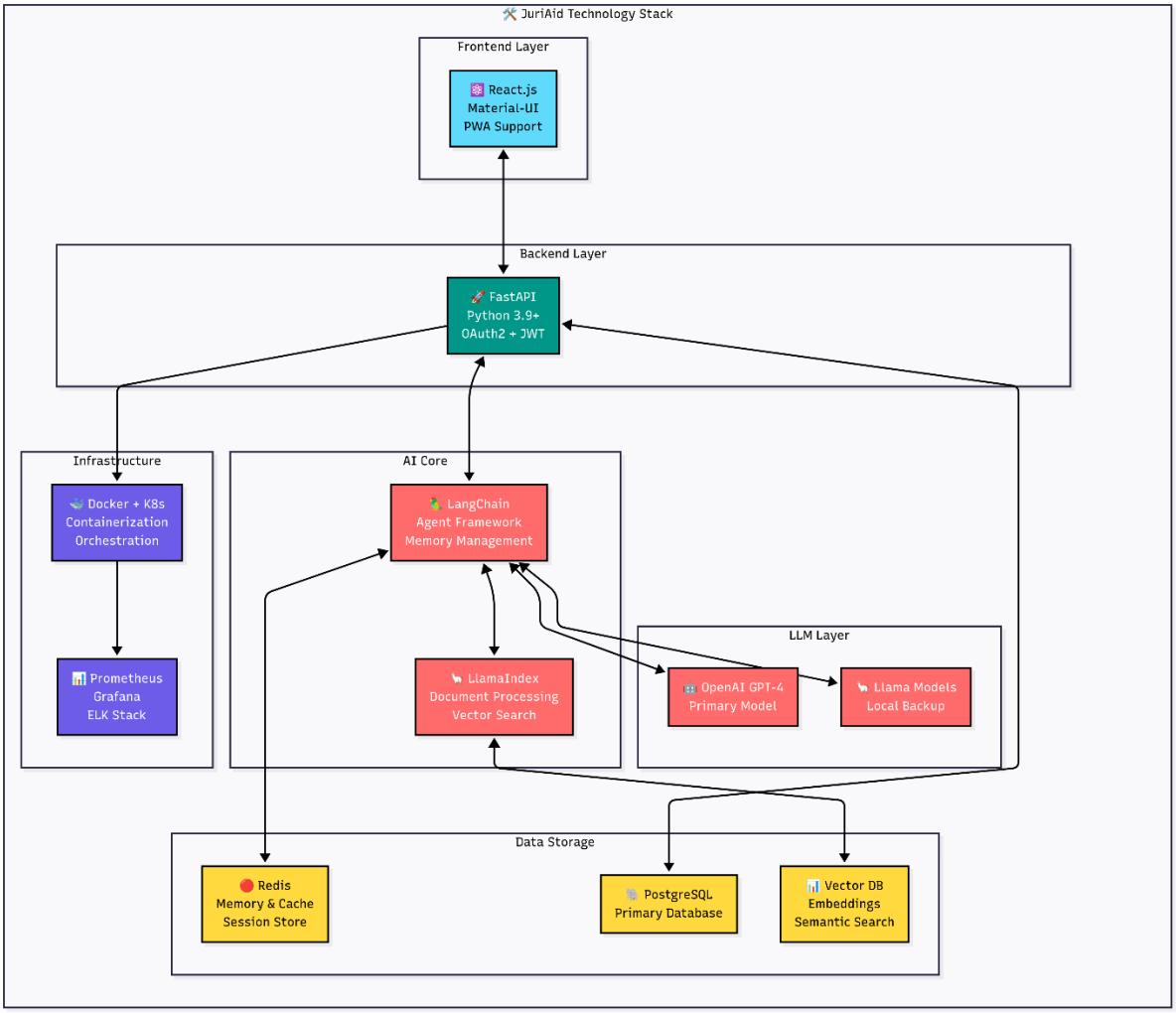

Figure 3 : Component technology diagram

4. Project Requirements

4.1.Functional Requirements

Functional requirements outline the precise actions, duties, or responsibilities that the system must carry out. For the Agentic AI Framework component, these include:

- **Query Processing:** The system shall accept and correctly interpret legal queries submitted in natural language English.
- **Task Decomposition:** The system shall analyze the user's query and autonomously decompose it into a logical sequence of executable sub-tasks.
- **Intelligent Agent Delegation:** The system shall accurately identify and delegate each sub-task to the appropriate specialist AI agent via a REST API call.
- **Stateful Context Management:** The system shall maintain the complete conversational history and contextual state for each unique legal case.
- **Response Synthesis:** The system shall be capable of aggregating the individual responses from all specialist agents and synthesizing them into a single, coherent final output.
- **Error Management:** The system shall detect and handle errors or timeouts from subordinate agents.

4.2.Non-Functional Requirements

Non-functional requirements define the criteria that describe the system's behavior and quality attributes.

- **Security:** All data in transit and at rest must be encrypted, complying with Sri Lanka's Personal Data Protection Act.
- **Reliability:** The Agentic AI Framework must be highly reliable, with a target uptime of 99.9%, and resilient to failures in subordinate agents.
- **Efficiency:** The end-to-end processing time for a standard legal query should not exceed 60 seconds.

- **Scalability:** The architecture must be designed for horizontal scalability and be deployable on cloud infrastructure.
- **Usability:** The API design must be clear, consistent, and well-documented to ensure easy integration for other team members.

5. Budget

The following amounts are estimated and may be subject to change:

Table 2: Budget Table

Description	Cost	Occurring
Server Hosting	Covered by GitHub Pack	Monthly
Cloud Database (e.g., AWS RDS for legal data storage)	Covered by GitHub Pack	Monthly
Legal Metirials	LKR 10,000	Annual
Developer Tools/Software Subscriptions	LKR 15,000 (Estimated Value)	Annual
LLM API Usage	LKR 12,000	Monthly

6. Commercialization

It is advised to launch this system as a **stand-alone entity** rather than being bound to a cooperative organization, especially given the current lack of a centralized, efficient platform in Sri Lanka that connects legal professionals with such AI tools. This approach ensures that any lawyer, regardless of their firm or affiliation, can utilize the system.

The system will be designed for **multi-device support**, with a dedicated **Mobile application for lawyers** to access its features. It will likely adopt a **freemium business model**. A free tier could offer basic question generation functionalities or a limited number of questions per case. The premium tier would unlock advanced features crucial for comprehensive legal practice, such as:

• **Unlimited, sophisticated question generation** with deeper legal context and nuance.

• **Integration with broader legal knowledge bases** and real-time legal updates.

• **Advanced fact elicitation and iterative refinement capabilities.**

7. Grant chart

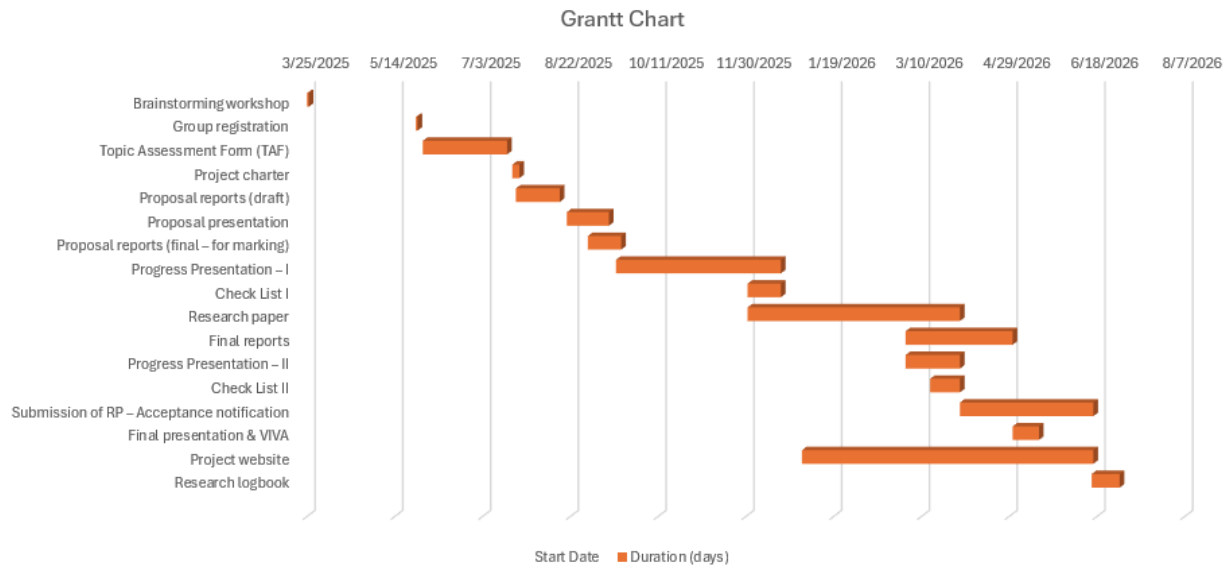

Figure 4 : Grant chart

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APPENDICES

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